

3 (a) (power =) work done / time (taken) or rate of work done A1 [1]

(b) (i) $F - R = ma$ C1

$$F = 1500 \times 0.82 + 1200$$

C1

$$= 2400 \text{ (2430) N}$$

A1 [3]

(ii) $P = Fv$ C1

$$= (2430 \times 22) = 53\,000 \text{ (53\,500) W}$$

A1 [2]

(c) (there is maximum power from car and) resistive force = force produced by car hence no acceleration

or

suggestion in terms of power produced by car and power wasted to overcome resistive force

B1 [1]